

AZ-900: Microsoft Azure Fundamentals

This ultimate guide includes all standard concepts plus highly specific differentiators, tricky definitions, and edge cases frequently found in actual AZ-900 exam questions.

1. Cloud Concepts & Shared Responsibility

Key Cloud Characteristics

- **High Availability (HA):** Uptime. Achieved through redundancy (Availability Zones/Sets).
- **Scalability:** *Vertical* (scaling up = more RAM/CPU). *Horizontal* (scaling out = adding more VMs).
- **Elasticity:** Automatically and dynamically adapting to workload changes via auto-scaling.
- **Reliability:** System recovery from failures.
- **Predictability:** Performance predictability (auto-scaling) and Cost predictability (TCO/Pricing calculators).
- **Consumption-based model (OpEx):** Pay-as-you-go. No upfront hardware costs (CapEx).

Cloud Model	Exam Differentiator
Public Cloud	Owned by Microsoft. Shared hardware. No CapEx. Highly scalable.
Private Cloud	Exclusive to one org. High CapEx. Complete control over security.
Hybrid Cloud	Combines Public/Private. Used for <i>Cloud Bursting</i> or strict data compliance.

Shared Responsibility Model

- **On-Premises:** You manage EVERYTHING.
- **IaaS (VMs, VNets):** Microsoft manages physical hardware. You manage OS, patching, apps, data.
- **PaaS (SQL DB, App Service):** Microsoft manages OS, hardware, runtime. You manage apps and data.
- **SaaS (M365):** Microsoft manages almost everything. You just manage your data and access.
- **Always Your Responsibility:** Your Data, Endpoints (Devices), Accounts, and Identities.

2. Azure Architecture & Management

Physical & Logical Infrastructure

- **Region:** A geographical area with networked datacenters. Some services are region-specific, some are global (like Entra ID, Front Door).
- **Availability Zone (AZ):** Physically separate datacenters *within* a region. Protects against local failures (fire, flood). *Not all regions have AZs.*
- **Region Pair:** Two regions in the same geography >300 miles apart. Platform updates are rolled out to paired regions sequentially, not simultaneously.
- **Subscriptions:** Billing and access control boundary. A user can have multiple subscriptions.
- **Resource Groups (RG):** Logical container. Resources in an RG can be in different regions. Deleting an RG deletes all resources inside. You cannot rename an RG.

Management Tools

- **Azure Portal:** Web-based graphical UI.
- **Azure CLI / Azure PowerShell:** Command-line tools for automation. Cross-platform.
- **Azure Cloud Shell:** Browser-based command line. *Requires an Azure Storage Account to function.*
- **Azure Arc:** Manage on-premises, AWS, or GCP servers directly from the Azure Portal.
- **Azure Resource Manager (ARM):** The deployment engine. Uses JSON templates, Bicep, or Terraform for Infrastructure as Code (IaC).

3. Core Compute, Storage & Networking Services

Compute Differentiators

- **Virtual Machines (VMs):** IaaS. Used for "lift and shift" (rehosting) legacy apps.
- **Azure Spot VMs:** Access unused Azure capacity at massive discounts. *Can be evicted at any time.* Perfect for batch processing or non-critical workloads.
- **Virtual Machine Scale Sets (VMSS):** Auto-scales identical VMs.
- **Azure App Service:** PaaS for web apps. Supports Windows/Linux.
- **Azure Functions:** Serverless. Event-driven. Billed per execution.
- **Azure Container Instances (ACI):** Fastest way to run a Docker container in Azure without managing VMs.
- **Azure Kubernetes Service (AKS):** Orchestrates large numbers of containers.
- **Azure Virtual Desktop (AVD):** Delivers Windows 10/11 desktops to any device globally.

Storage Types & Redundancy

- **Blob Storage:** Unstructured data (images, backups, logs).
- **Azure Files:** SMB/NFS file shares. Can be synced on-prem using *Azure File Sync*.
- **LRS (Locally Redundant):** 3 copies in ONE datacenter. Cheapest.
- **ZRS (Zone-Redundant):** 3 copies across 3 AZs in ONE region. Protects against datacenter failure.
- **GRS (Geo-Redundant):** Replicates to a paired region. Best for disaster recovery.

- **Storage Tiers:** Hot (frequent access, highest storage cost/lowest read cost), Cool (infrequent, >30 days), Archive (rare, >180 days, takes hours to retrieve, cheapest storage).

Networking Services

- **Virtual Network (VNet):** Private network. Can connect to on-prem via *VPN Gateway* (internet) or *ExpressRoute* (private, dedicated fiber connection).
- **VNet Peering:** Connects two VNets together seamlessly.
- **Network Security Groups (NSG):** Basic firewall for VNets. Filters traffic by IP/Port.
- **Azure Bastion:** Secure RDP/SSH access to VMs directly from the Azure Portal *without* exposing a public IP address on the VM.
- **Application Gateway:** Layer 7 load balancer with a Web Application Firewall (WAF).
- **Azure DNS:** Hosts DNS domains (does not register domain names, just hosts them).

4. Data, AI, & Developer Services

Databases & Big Data (Frequent Exam Targets)

- **Azure Cosmos DB:** Globally distributed, NoSQL database. Single-digit millisecond latency.
- **Azure SQL Managed Instance:** Easiest path to migrate legacy on-prem SQL Server to Azure (PaaS) without changing the app.
- **Azure Synapse Analytics:** Enterprise data warehousing and Big Data analytics.
- **Azure Databricks:** Apache Spark-based analytics platform optimized for Azure.
- **Azure HDInsight:** Open-source analytics (Hadoop, Kafka) in the cloud.

IoT & Developer Tools

- **IoT Hub:** PaaS backend to connect and monitor billions of IoT devices.
- **IoT Central:** SaaS solution/dashboard. Ready-made UI to connect and manage IoT devices quickly.
- **Azure Sphere:** End-to-end secure IoT solution (Hardware MCU + OS + Cloud Security Service).
- **Azure DevOps vs GitHub:** Both offer CI/CD pipelines and repos. DevOps is Microsoft's enterprise suite; GitHub is open-source heavy.
- **Azure DevTest Labs:** Quickly provision dev/test environments while enforcing strict cost limits and auto-shutdown rules.

5. Identity, Governance, & Security

Identity (Microsoft Entra ID)

- **Entra ID (formerly Azure AD):** Cloud IAM. Not the same as Windows Server Active Directory (uses HTTP/REST, not Kerberos/LDAP).
- **B2B vs B2C:** *B2B (Business-to-Business)* lets guest users from other orgs access your resources. *B2C (Business-to-Consumer)* lets customers log in using social accounts (Google, Facebook).
- **Conditional Access:** Enforces rules (e.g., "Require MFA if user is outside the country" or "Block access from risky IPs").
- **RBAC (Role-Based Access Control):** Controls *who* can do what (e.g., Virtual Machine Contributor).

Governance: Policy vs. RBAC vs. Locks

- **Azure Policy:** Controls *what* you can create (e.g., "Only allow creating VMs in East US", "Require a 'CostCenter' tag"). Evaluates compliance.
- **Azure RBAC:** Controls *who* has access to resources.
- **Resource Locks:** Prevents accidental deletion. *CanNotDelete* (can read/modify, but not delete). *ReadOnly* (cannot modify or delete). Locks override RBAC (even an Owner cannot delete a locked resource).
- **Microsoft Purview:** Discovers and classifies sensitive data across your entire organization.

Security & Monitoring

- **Defender for Cloud:** Provides a *Secure Score* and automated security recommendations. Includes Just-In-Time (JIT) VM access.
- **Microsoft Sentinel:** SIEM/SOAR system. Analyzes logs across the entire enterprise to hunt for threats.
- **Azure Key Vault:** Securely stores passwords, API keys, and certificates.
- **Azure Monitor:** Collects logs and metrics. Features include *Application Insights* (web app performance) and *Log Analytics*.
- **Azure Service Health:** Alerts you to Azure regional outages or planned maintenance affecting *your* specific deployed resources.

6. Cost Management & Support Plans

Cost Tools (Know the exact difference)

- **Pricing Calculator:** Estimate costs *before* you provision new resources.
- **TCO Calculator (Total Cost of Ownership):** Compares running workloads on-prem vs. migrating to Azure over a 3-5 year period.
- **Cost Management & Billing:** Track *historical/current* spending, set budgets, and create billing alerts for deployed resources.
- **Tags:** Metadata (Key:Value) used to group resources for billing/cost allocation across different departments.

Support Plans (Highly Tested)

Plan	Key Features
Basic (Free)	Billing and subscription support only. NO technical support. Includes Azure Advisor.
Developer	Business hours technical support via email. For non-production environments.
Standard	24/7 technical support by phone/email. Production workloads.
Professional Direct	24/7 support + Architecture guidance + Dedicated account manager.

SLAs (Service Level Agreements): A 99.9% SLA guarantees a certain uptime. If Microsoft fails to meet the SLA, you receive *Service Credits* (not a cash refund). Free tier services have NO SLA. Two services combined (e.g., 99.9% VM + 99.9% DB) result in a **Composite SLA** which is lower than the individual SLAs (you multiply them).